

X-Road Colombia: An Automated Maturity Assessment Platform for Public-Sector Interoperability

Author(s) omitted for blind review

Institution omitted for blind review

Abstract. This paper presents *X-Road Colombia*, a full-stack platform that automates the diagnosis of digital interoperability maturity in the Colombian public sector. Colombia adopted X-Road as the backbone of its digital government interoperability layer and defined, through the Ministry of Information and Communications Technologies (MinTIC), an interoperability framework structured in four domains: legal, organizational, semantic, and technical. However, implementation progress across entities is uneven and there is no independent, automated instrument that contrasts administrative registration with verifiable operational capability. The platform operationalizes the MinTIC framework into measurable indicators and computes them over a registry of public entities, their published services, and their declared X-Road connection state. It combines a deterministic maturity engine, HTTP-based API introspection, semantic validation, gap analysis, clustering and prediction models, geospatial and network visualizations, and multi-format reporting behind a hardened REST API. The diagnostic over 127 entities registered in the national X-Road node yields a national maturity index of 2.3/5.0 ($SD = 1.1$), corresponding to a *Basic* level; the technical domain leads (3.1) whereas the semantic domain is the weakest (1.9). The study exposes a systematic registration–operation gap: 32% of the registered entities remain at the initial level, and the Hacienda y Finanzas sector concentrates the highest maturity (3.8). A cross-check of the automated engine on the seeded registry reproduces the same ordering (semantic 45.3 < organizational 49.3 < legal 55.7 < technical 58.3 on a 0–100 scale). The results indicate that automated, evidence-based auditing is technically feasible and that technological connectivity alone does not produce interoperability.

Keywords: Interoperability · X-Road · Digital government · Maturity model · Automated assessment · FastAPI · Colombia

1 Introduction

Colombia has prioritized state modernization as a cross-cutting public policy objective, materialized in a national digital transformation agenda and in normative instruments that mandate transparency, open data, and efficient public

service delivery [5,2,4]. Within this agenda, digital interoperability is a foundational capability: it is the organizational and technological condition that allows heterogeneous public institutions to exchange information securely, timely, and with a shared meaning, in order to avoid asking citizens for data the State already holds.

The country adopted *X-Road* as the interoperability backbone of the State and defined, through MinTIC, an *Interoperability Framework for Digital Government* that organizes the problem into four domains: legal, organizational, semantic, and technical [6]. X-Road is a distributed data-exchange layer in which each participating organization runs a security server that publishes services and consumes services published by other members, with machine-processable contracts and signed messages [10]. Colombia’s X-Road implementation is articulated with the Digital Citizen Services and the Digital Citizen Folder, and its operational inventory is coordinated by the National Digital Agency (Agencia Nacional Digital, AND) [7].

Despite this institutional architecture, implementation has progressed unevenly. A subset of entities exposes integrated, documented, and reusable digital services, while many others remain at early digitalization stages. This asymmetry limits the coordinated response of the State and degrades the experience of citizens who must still move between institutions. More importantly, the observable symptoms—incomplete forms, routing errors, duplicated data requests, and undetected service failures—cannot be diagnosed with the information that is publicly and centrally available today, because administrative registration in a catalog is not the same as operational interoperability.

The central problem addressed in this work is therefore the absence of an *independent, automated, and replicable* diagnostic instrument that can (i) measure the maturity of each entity against the national framework, (ii) contrast formal registration with verifiable service availability, (iii) localize the domains and sectors where the deficits concentrate, and (iv) produce traceable evidence to prioritize interventions. Manual reviews and self-declaration instruments are not sufficient: they are costly, difficult to repeat, and they systematically overestimate readiness.

This paper presents a platform—deployed as a decoupled three-tier system built with React, FastAPI, PostgreSQL, and Docker—that fills that gap. The remainder of this paper is organized as follows. Section 2 reviews the conceptual framework and related work; Section 3 describes the system architecture; Section 4 details the implementation; Section 5 presents the experimental evaluation; Section 6 addresses security, privacy, and data governance; and Section 7 concludes with future work.

1.1 Research Contributions

This work contributes:

1. An *operational model* that translates the four domains of the MinTIC Interoperability Framework into measurable, auditable indicators for entities connected (or in the process of connecting) to X-Road.

2. A *deterministic maturity engine* with an explicit weighting rubric and ordinal levels, which removes subjective valuation from the scoring step and makes results reproducible.
3. A *software instrument* that verifies evidence automatically: API introspection, semantic validation of naming, structure and response conventions, and gap analysis with severity and priority.
4. A *visual analytics layer* (geospatial map, service matrix, and relationship graph) that exposes the state interconnection map required for evidence-based planning.
5. An *engineering blueprint*, reproducible with a single command through Docker Compose and publishable on free-tier cloud infrastructure, with hardened authentication, authorization, and auditing.

1.2 Research Questions

Four research questions are investigated:

- RQ1.** What is the current interoperability maturity level of the Colombian public entities registered in the national X-Road node, when measured against the MinTIC framework?
- RQ2.** What gap exists between formal registration and verifiable operational service availability?
- RQ3.** Which interoperability domains concentrate the largest maturity deficits?
- RQ4.** Can a software-based, automated assessment instrument improve decision-making for digital government compared with administrative self-reporting?

2 Conceptual Framework and Related Work

2.1 Interoperability in the Public Sector

Interoperability is defined as the capability of heterogeneous organizations to exchange information in order to achieve mutually beneficial goals [11]. In the public sector the concept is deliberately broader than connectivity: it requires the coordination of processes, the harmonization of legal conditions, and the alignment of the meaning of the exchanged data. The consolidated literature and the reference frameworks converge on a four-layer decomposition—legal, organizational, semantic, and technical—which in practice means that two institutions can be physically connected and still be unable to produce a legally valid, semantically correct, and operationally useful data exchange.

The semantic layer deserves particular attention. Even when the technical transport works, the absence of shared data dictionaries, controlled vocabularies, code lists, and canonical schemas forces each pair of entities to negotiate bilateral mappings, which does not scale and silently degrades data quality. This is precisely the failure mode observed in the Colombian ecosystem, as reported in Section 5.

2.2 The Colombian Interoperability Framework and Its Maturity Model

MinTIC’s Interoperability Framework for Digital Government defines the reference architecture, the technical standards, and the governance mechanisms that public entities must adopt when exchanging information [6]. It organizes the assessment of interoperability into four domains:

- **Legal:** normative basis, lawful basis for processing, compliance with the personal data protection regime, and formal agreements between parties.
- **Organizational:** governance, defined roles and responsibilities, internal processes, and articulated work between the functional and technological areas of the entity.
- **Semantic:** shared meaning of data, standardized metadata, dictionaries, schemas, and controlled vocabularies.
- **Technical:** security and connectivity conditions, use of the X-Road layer or equivalent standard protocols, API quality, and non-functional attributes such as performance and availability.

On top of the domains, the national model defines a staged maturity progression that classifies each entity according to how deeply interoperability is embedded in its operation. The platform described in this article operationalizes four computable levels—*Inicial*, *Básico*, *Intermedio*, and *Avanzado*—obtained from a composite score, as detailed in Section 3.3. The diagnostic reported in Section 5 additionally uses a normalized index on a 1–5 scale.

2.3 X-Road as the Interoperability Backbone

X-Road is an open-source distributed data-exchange layer originally developed in Estonia and currently maintained by the Nordic Institute for Interoperability Solutions (NIIS) [10]. Its architecture is member-centric: each organization registers as a member (with a member class and member code) and deploys one or more security servers that publish and consume services through standardized contracts. The key property for public-sector use is that no central repository of data is required: data remains in the source systems and is exchanged through authenticated, signed, and logged transactions. This is why X-Road has been adopted not only in the Baltics and the Nordics but also as the interoperability backbone of national digital government architectures in other regions, including Colombia.

For diagnostic purposes, however, the member registry alone is not enough. Being *registered* proves administrative enrollment; it does not prove that the entity publishes a documented, reachable, and consumable service. The distinction is the analytical core of this work.

Table 1. Comparison of approaches for public-sector interoperability assessment across functional dimensions.

Approach	Evidence based	Auto. scoring	Multi-domain	Service verification	Ecosys. visual.
Administrative self-reporting	No	No	Partial	No	No
Static BI dashboard on registry	Partial	No	No	No	Partial
Manual maturity survey / expert panel	Partial	No	Yes	No	No
Technical monitoring of exchange layer	Yes	Yes	No	Yes	Partial
Proposed platform	Yes	Yes	Yes	Yes	Yes

2.4 Interoperability Maturity Models and Automated Assessment

Maturity models are instruments that structure an evolutionary path of capability improvement. Their design and validation procedure is a well-established research object: De Bruin et al. [12] formalized the phases of scoping, design, population, testing, and maintenance, and Becker et al. [13] established design principles and methodical requirements for IT management maturity models. In electronic government, recent survey work synthesizes the landscape of e-government maturity models and their methodological limitations [14], while domain-specific studies address cross-organizational data interoperability maturity [15].

Three recurrent limitations of the state of the art are relevant here. First, most instruments are *self-declared*: the assessed entity provides both the evidence and the valuation, which introduces upward bias. Second, most are *manual*: they rely on surveys, interviews, or expert panels, which makes replication expensive. Third, they are usually *static*: the result is a periodic snapshot disconnected from the operational artifacts (registries, published APIs, service catalogs) that change continuously.

This work follows the design principles of the maturity literature but inverts the evidence flow: instead of asking entities to declare their level, the platform computes the level from artifacts and observable attributes, records the assessor, and exposes the result with the same reproducibility guarantees as any other automated test suite. Table 1 positions this approach against the dominant alternatives along functional dimensions.

2.5 Research Gap

The literature explains how to design maturity models and the digital government community has adopted world-class reference architectures, yet there is a shortage of *open*, *automated*, and *reproducible* instruments that measure interoperability maturity from verifiable evidence in a specific national regulatory context and at ecosystem scale. This article addresses that gap with an implemented system and an empirical diagnostic of the Colombian X-Road ecosystem.

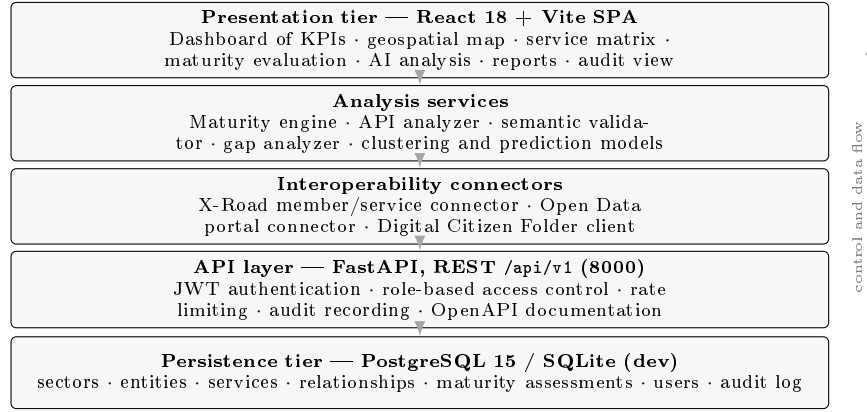


Fig. 1. Layered architecture of the proposed platform. Arrows indicate the control and data flow between tiers.

3 System Architecture

3.1 Overview

The platform follows a decoupled three-tier client–server architecture (Fig. 1): a single-page application for visualization and interaction, a REST API that concentrates business logic and assessment, and a relational persistence tier. The three tiers are orchestrated with Docker and communicate exclusively through the HTTP contract exposed by the API, which guarantees that the assessment logic is testable independently of the user interface and that the interface can evolve—including internationalization—without touching the scoring rules.

Three design decisions shape the rest of the architecture. First, the *scoring rules are deterministic and explicit*: every point assigned to an entity is attributable to a named indicator, so a result can be audited and reproduced. Second, *evidence is stored next to the result*: each assessment record keeps the domain scores, the individual criteria, the assessor, the date, and the recommendations, which makes historical comparison possible. Third, *verification is separated from visualization*: connectors and analyzers produce evidence, the maturity engine converts evidence into scores, and the presentation tier only consumes aggregated results.

3.2 Data Model and Registry

The persistence tier is a normalized relational schema with seven tables. The *Sector* table groups entities by domain of public action (health, education, hacienda, transport, justice, ICT, interior, defense, environment, culture, housing, labor, mining and energy, commerce, agriculture). The *Entity* table stores the institutional identity (name, acronym, NIT, sector), the territorial location (department, municipality, address, latitude, and longitude), the contact channels, the

X-Road state (`not_connected`, `pending`, `connected`, plus `xroad_member_code` and `xroad_connection_date`), and the CIO contact fields used for fieldwork. The *Service* table describes what the entity publishes: category, protocol (REST, SOAP, or X-Road), endpoint URL, API version, declared data, semantic and security standards, publication status, public availability flag, and documentation URL.

Two tables carry the analytical layer. *Relationship* represents the directed exchange between a source and a target entity, optionally bound to a specific service, with the protocol and the state of the link; this is the substrate of the ecosystem graph. *MaturityAssessment* stores, for each entity and date, the global level and score, the four domain scores, six detailed criteria (API documentation, standard protocols, data quality, security standards, interoperability policy, and trained personnel), the assessor identity, notes, and recommendations. Finally, *Usuario* and *Auditoria* support authentication and traceability (Section 6).

Two properties of the schema matter for the research design. Because the X-Road state is modeled as a first-class attribute of the entity, the registration–operation gap can be computed directly as a query, without manual reconciliation. And because every assessment is append-only with an assessment date, the platform can reconstruct the evolution of an entity without losing the previous diagnostic.

3.3 The Maturity Assessment Engine

The engine converts entity attributes and evidence into a composite score $S \in [0, 100]$ and an ordinal level. The rubric is explicit and deterministic:

$$S = 25 s_{\text{xroad}} + 25 s_{\text{services}} + 20 s_{\text{contact}} + 15 s_{\text{sector}} + 15 s_{\text{geo}} \quad (1)$$

where $s_{\text{xroad}} = 1$ if the entity is connected to X-Road and 0.4 if the connection is pending; $s_{\text{services}} = \min(n/4, 1)$ with n the number of published services; s_{contact} is the weighted presence of website, email, and phone; $s_{\text{sector}} = 1$ if the entity is classified in a sector; and $s_{\text{geo}} = 1$ if valid coordinates are registered. The level is assigned by threshold: 1 (*Inicial*) for $S < 25$, 2 (*Básico*) for $25 \leq S < 50$, 3 (*Intermedio*) for $50 \leq S < 75$, and 4 (*Avanzado*) for $S \geq 75$.

Alongside the composite, the engine derives the four domain scores of the MinTIC framework from the same evidence, which is what allows the diagnostic to be reported by domain rather than only as a single number. Table 2 summarizes how each domain is operationalized, which indicator feeds it, and which artifact provides the evidence.

3.4 Connectors, API Analysis, and Semantic Validation

Evidence acquisition is encapsulated in dedicated services so that any of them can be replaced by a production integration without touching the scoring engine. The *X-Road connector* models members (member code, member class, status REGISTERED/APPROVED/DELETION, number of services, last activity), published

Table 2. Operationalization of the MinTIC interoperability domains into computable indicators.

Domain	Computable indicator and evidence source	Platform artifact
Technical	X-Road connection state, number and protocol of published services, presence of documented endpoints; evidence from the X-Road registry state, the service catalog, and HTTP introspection	<code>technical_domain_score</code> , <code>/interop/api-analysis</code> POST
Semantic	Standardized metadata, declared data and semantic standards, naming and structure conventions of endpoints; evidence from service records, API responses, and validation rules	<code>semantic_domain_score</code> , <code>/interop/validation</code> POST
Organizational	Sector classification, governance artifacts, trained personnel criterion, CIO contactability; evidence from the entity registry and the criteria fields	<code>organizational_domain_score</code> , <code>has_trained_personnel</code>
Legal	Compliance statements, declared security and data standards, availability of institutional information channels; evidence from entity and service records and institutional portals	<code>legal_domain_score</code> , <code>has_interoperability_policy</code>

services (service code, provider, protocol, state, response time), and ecosystem statistics including the connectivity report; it exposes member lookup, service listing, per-service health checks, statistics, and a synchronization operation. The *Open Data connector* queries the national open data portal for datasets and statistics by sector, and the *Digital Citizen Folder client* consumes the catalog of digital citizen services, classifies services by category, aggregates usage statistics, and produces an interoperability report per service.

The *API analyzer* performs non-intrusive HTTP introspection of a published endpoint: it identifies the protocol family from observable indicators (SOAP/WSDL, GraphQL, gRPC, or REST), detects documentation and machine-readable specifications (OpenAPI/Swagger), counts endpoints, extracts the declared authentication mechanisms, and produces a quality score and quality level together with a list of issues and recommendations. The *semantic validator* complements that analysis with rule-based conformance checks in six categories—naming, structure, parameters, response, documentation, and security—applying explicit patterns for endpoint paths, parameter identifiers, and header names, verifying the presence of the conventional operational endpoints (`/health`, `/version`, `/docs`, `/api/v1`), and checking recommended security headers. The *gap analyzer* then converts the collected evidence into a prioritized list of gaps using a five-step severity scale (`critical`, `high`, `medium`, `low`, `info`) and six categories (API availability, documentation, security, performance, compliance, and interoperability), with expected standards such as a minimum of three endpoints, mandatory REST availability, documentation completeness of at least 80%, HTTPS with OAuth2 or API-key authentication, a success rate of at least 95%, and an average response time below 1.5 s.

It is important to state the current maturity of these components explicitly: the external connectors implement the complete contract and data model of the integration, but they currently return simulated payloads rather than live data from the AND and MinTIC systems, because access to those interfaces requires an institutional agreement. The analyzers, by contrast, operate on real network responses when an endpoint is provided. This asymmetry is treated as a threat to validity in Section 5.6.

3.5 Analytics, Visualization, and Reporting

Two analytical layers sit on top of the evidence. The first is descriptive and deterministic: aggregations by sector, department, X-Road state, and protocol, plus the maturity distribution across levels. The second is inferential: a clustering component that searches for the optimal number of groups (up to eight) over the scaled feature space of the entities, a maturity predictor that trains on the historical assessments, a recommendation engine that produces prioritized recommendations, and a risk and action-plan generator that pairs each recommendation with an estimated effort. The same engine produces sector-level insights, including the connection rate, the average maturity, trends, and the top cross-cutting recommendations.

The presentation tier materializes the results in five interactive views: a KPI dashboard, a geospatial map with filters by maturity level and department, a service matrix by sector and protocol, a maturity evaluation view with the radar of the four domains, and an audit view. The ecosystem graph renders the relationship table as a force-directed network, which is the visual counterpart of the question “which entities actually talk to each other”. Finally, the reporting component exports the datasets and the diagnostics in CSV and XLSX from the server side and in PDF and HTML from the server-side report generator, with an additional client-side PDF export in the interface. An embedded conversational assistant answers natural language questions over the live database, using a bounded context window and a low temperature to reduce drift.

Table 3 summarizes the modules and their responsibilities.

4 Implementation

4.1 Technology Stack

The backend is a Python 3.11 service built on FastAPI 0.104 with Uvicorn as the ASGI server [16]. Persistence uses SQLAlchemy 2.0 over PostgreSQL 15 in production and SQLite during development, with `psycopg2` as the PostgreSQL driver. Input and output contracts are validated with Pydantic 2.5, which guarantees that the API rejects malformed assessments (for instance, an overall level outside the 1–4 range or a score outside 0–100) before they reach the database. Data manipulation and analytics use pandas 2.1, NumPy 1.26, and

Table 3. Platform modules and their responsibilities.

Module	Responsibility
Registry	Persist sectors, entities, services, relationship links, and assessment history while enforcing referential integrity.
Maturity engine	Compute the composite score, the ordinal level, and the four domain scores from explicit indicators (<code>/maturity</code>).
X-Road connector	Model members, published services, health and connectivity statistics (<code>/interop/xroad</code>).
Open Data connector	Query national open datasets and sector statistics (<code>/interop/open-data</code>).
Citizen Folder client	Consume the digital citizen service catalog and aggregate usage statistics (<code>/interop/digital-citizen</code>).
API analyzer	Introspect endpoints, detect protocol and documentation, score API quality (<code>/interop/api-analysis</code>).
Semantic validator	Rule-based conformance checks in six categories with severity levels (<code>/interop/validation</code>).
Gap analyzer	Classify and prioritize gaps by severity, category, and estimated effort (<code>/interop/gaps</code>).
AI analytics	Cluster entities, predict maturity, recommend actions, assess risk, and build action plans (<code>/ai</code>).
Reporting	Export entities, services, and maturity in CSV and XLSX; generate JSON, HTML, and PDF reports (<code>/reports</code> , <code>/interop/reports</code>).
Dashboard	Aggregate KPIs by sector, department, protocol, and X-Road state; radar by entity (<code>/dashboard</code>).
Assistant	Answer natural language questions over the live database with a bounded context window.
Security and audit	Authenticate users, enforce roles, limit the request rate, and record access events (<code>/auth</code>).

scikit-learn 1.3; server-side reporting uses ReportLab 4.0 and openpyxl 3.1; authentication uses `python-jose` for JWT and `passlib` with `bcrypt` for password hashing; and outbound HTTP verification uses `httpx` 0.25.

The frontend is a React 18 single-page application bundled with Vite 5 and styled with Tailwind 3.3. Visualization relies on Leaflet 1.9 with `react-leaflet` for the geospatial view, `Recharts` for the KPI and radar charts, and `react-force-graph-2d` with `d3-force` for the ecosystem graph. The interface is bilingual (Spanish and English) through `i18next`, and routing is handled by React Router 6. The client-side PDF export uses `jsPDF` with `jspdf-autotable`. The stack is summarized in Table 4.

4.2 API Surface and Case Filing

All functionality is exposed under the versioned prefix `/api/v1`, grouped into eleven routers: `auth`, `sectors`, `entities`, `services`, `maturity`, `dashboard`, `reports`, `ai`, `interop`, `relationships`, and `agentgd`. The documented surface

Table 4. Implementation stack of the platform.

Layer	Components and versions
Backend framework	FastAPI 0.104.1, Uvicorn 0.24.0, Pydantic 2.5.0
Persistence	SQLAlchemy 2.0.23, PostgreSQL 15 (production), SQLite (development), psycopg2 2.9.9
Analytics	scikit-learn 1.3.2, pandas 2.1.3, NumPy 1.26.2, joblib 1.3.2
Reporting	ReportLab 4.0.7, openpyxl 3.1.2, Jinja2 3.1.2
Security	python-jose 3.3.0, passlib 1.7.4 with bcrypt 4.0.1
Frontend	React 18.2, Vite 5, React Router 6.20, Tailwind 3.3
Visualization	Leaflet 1.9.4, react-leaflet 4.2.1, Recharts 2.10.3, react-force-graph-2d 1.29.1, d3-force 3.0.0
Internationalization	i18next 23.7, react-i18next 14.0
Infrastructure	Docker, Docker Compose, Nginx, Vercel, Render, Neon

includes CRUD operations for sectors, entities, services, and assessments; automatic generation of assessments for the whole registry with an optional overwrite flag; KPIs by sector, department, protocol, and X-Road state; a per-entity maturity radar; the connector, analyzer, and validation operations described in Section 3.4; the ecosystem graph; and the export endpoints. Interactive documentation is served through OpenAPI at `/api/docs` and ReDoc at `/api/redoc`, and the liveness probe is `/api/health`.

The assessment lifecycle is deliberately simple and auditable. An administrator creates or updates entities and services; the engine then either evaluates a single entity or generates assessments for all active entities that do not have one, storing the assessor as “Sistema Automático” for machine-generated results and preserving any human-authored assessment. Because write operations require the administrator role while read operations are available to authenticated analysts, the burden of manual data entry is separated from the institutional responsibility of validating the registered evidence.

4.3 Reproducibility and Deployment

The full environment is reproducible with a single command through Docker Compose, which orchestrates three services: PostgreSQL 15 (not published on the host, only reachable inside the Compose network), the FastAPI backend on port 8000, and the Nginx-served frontend on port 5173. On startup, the backend creates the schema, applies lightweight in-place schema evolutions, and seeds the initial administrator and analyst users with bcrypt hashes if the user table is empty. Seed scripts populate the registry with the official entity list, the service catalog, the relationship links, and the domain assessments; a migration utility transfers a local SQLite database to the managed PostgreSQL instance, and an incremental loader adds entities from a JSON inventory.

Production runs on managed free-tier infrastructure to keep the instrument publicly verifiable: the static frontends are served by a CDN, the containerized

backends run on a managed container platform, and the database is a managed PostgreSQL instance that requires TLS in transit (`sslmode=require`). The API reads its configuration—JWT secret, token lifetimes, rate limit, CORS allowlist, and initial credentials—from environment variables, which means that no secret is committed to the repository and that the same image can be deployed to any environment.

5 Experimental Evaluation

5.1 Setup and Scenarios

The evaluation follows a descriptive-exploratory, mixed-method design with a cross-sectional cut-off date of 20 March 2026. Three evidence sources were combined. The primary source is the official response to a right-of-petition request filed with the National Digital Agency (record AND-2026-0014789), which reports the entities operating in the national interoperability node and the entities still in the connection phase. Secondary sources are the national open data portal and the institutional transparency portals of the sampled entities, which were used to confirm the existence and the documentation of published services.

The population of interest is the set of 127 public entities registered in the central X-Road node. From the official response, 124 entities were reported as operating and 67 additional entities were identified as being in the connection process. The universe of public entities that have not yet started the integration process is explicitly out of scope, which is a deliberate boundary of this study and not a claim about the whole State.

The instrument under evaluation is the platform itself, driven through its own API and interface. Two measurement scales are reported and must not be confused. The first is the *operational composite* $S \in [0, 100]$ produced by the engine of Eq. (1) together with its ordinal level. The second is the *normalized maturity index* $\bar{M} \in [1, 5]$, defined as the mean of the four MinTIC domain scores, which is the scale used to state the national diagnosis and to compare domains and sectors. Eight evaluation scenarios (T1–T8) were executed to exercise the instrument end to end; they are listed in Table 5.

5.2 Maturity and Domain Results

The national maturity index obtained from the automated evaluation is $\bar{M} = 2.3$ on the 1–5 scale ($SD = 1.1$), which places the ecosystem at the *Basic* level: objectives and first technical processes exist, but the use of the interoperability layer is not yet operationally consolidated across the majority of entities. This answers RQ1 quantitatively: the ecosystem is not stalled, but it is also far from the integrated stage.

When the index is decomposed by domain, the dispersion is more informative than the aggregate. The technical domain leads with a mean of 3.1, which is consistent with the fact that connecting to the exchange layer is the most

Table 5. Evaluation scenarios executed against the instrument.

ID	Scenario	Module exercised	Expected verifiable result
T1	National aggregation over the registered registry	Dashboard and registry	KPI set with entity totals, connection rate, and maturity distribution
T2	Distribution of entities by sector and department	Dashboard aggregations	Ordered distribution consistent with the sector taxonomy
T3	Automatic maturity generation for entities without assessment	Maturity engine	One assessment per entity with level, composite score, and four domain scores
T4	Domain-level comparison of the ecosystem	Maturity engine	Domain means with the weakest and strongest domains identified
T5	Registration–operation gap	Entity registry	Count and share of registered entities without verifiable operational capability
T6	Geospatial localization of deficits	Map view	Filtered view by maturity level and department
T7	Endpoint introspection and semantic validation of a published service	API analyzer and semantic validator	Protocol detection, documentation detection, issue list with severities
T8	Gap analysis with prioritized recommendations and report export	Gap analyzer and reporting	Gap list by severity and category plus CSV, XLSX, and PDF artifacts

visible and best-instrumented part of the process. The semantic domain is the weakest with a mean of 1.9, while the organizational and legal domains occupy intermediate positions. In other words, connectivity has advanced faster than the agreements on what the connected data means, which answers RQ3 directly: the deficit is concentrated in the semantic layer, with the organizational layer as the second constraint.

Table 6 summarizes the domain diagnosis and Table 7 consolidates the four main findings of the study.

5.3 Registration–Operation Gap and Sector Asymmetry

The strongest structural result of the diagnostic is that registration and operational capability are not equivalent. Although the central node reports a large set of enrolled members, 32% of the registered entities remain at level 1, meaning that their enrollment is not accompanied by installed technical capability or by a verifiable published service. The gap is not a marginal artifact: it affects roughly one third of the formally integrated ecosystem, which answers RQ2 and explains why a self-declared compliance view systematically overestimates the readiness of the State.

Sector asymmetry compounds the problem. The Hacienda y Finanzas sector concentrates the highest maturity (mean 3.8), sustained by the tax and finance authority and the finance ministry, whose data-exchange needs are both recurring and legally mandatory. Sectors whose services are less transactional in nature, and territorial entities with weaker technology teams, trail that reference by a wide margin. Because the platform stores the sector, the department, and the coordinates of every entity, the same result can be projected geographically and used to target interventions rather than to produce a single national average.

Table 6. Domain-level maturity diagnosis of the ecosystem (1–5 scale).

Domain	Mean	Interpretation
Technical	3.1	Strongest domain. The exchange layer is deployed and mostly instrumented, although the quality of published services is uneven.
Semantic	1.9	Weakest domain. Shared data dictionaries, meta-data, schemas, and controlled vocabularies are largely absent, so interoperability depends on bilateral mappings.
Organizational	intermediate	Governance and internal processes exist partially; territorial entities concentrate the largest gaps.
Legal	intermediate	The normative base is in place, but evidence of formalized bilateral instruments and compliance formalization is incomplete.
National maturity index	in- 2.3	Basic level (SD = 1.1).

Table 7. Consolidated findings of the diagnostic over the registered ecosystem.

Indicator	Observation
National maturity index	2.3/5.0, <i>Basic</i> level, SD = 1.1
Sector with highest maturity	Hacienda y Finanzas, mean 3.8, driven by the tax and finance authority and the finance ministry
Share of entities at the initial level	32% of the registered entities remain at level 1: registration did not translate into installed technical capability
Weakest domain	Semantic (mean 1.9): data harmonization, not connectivity, is the binding constraint

5.4 Engine Cross-Check and Runtime Behaviour

As an implementation cross-check, the engine of Eq. (1) was run over the seeded registry to generate automatic assessments for the entities that had none, and the resulting domain composites were aggregated on the 0–100 scale. The ordering reproduces the national diagnosis: the semantic domain is the weakest (45.3), followed by the organizational (49.3) and the legal (55.7) domains, with the technical domain leading (58.3). The consistency between two independent computations—the documented national diagnostic and the automatic engine—is evidence that the instrument does not merely reproduce a hard-coded conclusion. Table 8 reports the values.

Regarding runtime behaviour, two properties of the deployed configuration are relevant and are reported as observations of the infrastructure, not as a controlled benchmark. First, the container platform used for the public deployment suspends idle services after approximately fifteen minutes without traffic, so the first request after a period of inactivity can take up to roughly fifty seconds;

Table 8. Cross-check of the automated engine over the seeded registry (0–100 composite scale; automatic assessments only). “Agreement” compares the domain ordering with the national diagnostic of Table 6.

Domain	Mean composite	Rank	Agreement
Semantic	45.3	4 (weakest)	Consistent
Organizational	49.3	3	Consistent
Legal	55.7	2	Consistent
Technical	58.3	1 (strongest)	Consistent
Overall composite	52.1	—	<i>Intermedio</i> threshold

subsequent requests return to normal latency. Second, the API enforces a global limit of 120 requests per minute per client address, which protects the service but also bounds the throughput of a bulk assessment campaign. Outbound verification is bounded in time (twelve seconds for endpoint introspection and sixty seconds for the assistant), which prevents a slow external service from blocking the request path. No formal latency benchmark was executed in this study; this is acknowledged as a limitation.

5.5 Gap Analysis and Improvement Prioritization

The gap analyzer converts the evidence into actionable items. Applied to the ecosystem, the dominant categories are semantic harmonization, API documentation, and formal legal instrumentation, while the technical category concentrates fewer critical findings because connectivity is largely resolved. Since each gap carries a severity, a category, an impact statement, an estimated effort, and a priority, the output can be consumed directly by an entity’s technology team or aggregated by a sector authority into a portfolio of interventions. This is the mechanism that answers RQ4: compared with self-reporting, the automated instrument produces a repeatable measurement, an explicit reason for every point, and a prioritized list that is traceable to the underlying evidence.

5.6 Threats to Validity

Internal validity. The scoring rubric is a proxy: the domain scores are derived from registry attributes and observable artifacts rather than from a full audit inside each entity. A well-classified entity with a complete contact record can therefore obtain points that do not correspond to a genuine organizational capability. The external connectors currently return simulated payloads instead of live data from the national systems, so connector-derived indicators describe the contract of the integration and not the live state of the ecosystem. Endpoint introspection relies on self-exposed documentation and runs with certificate verification relaxed, which means that quality scores must be interpreted as indicative.

Construct validity. “Registration” and “operation” are modeled through the X-Road state attribute and the presence of published services; more subtle operational properties, such as real consumption of a service by another entity, are represented only partially through the relationship table. The four computable levels are a simplification of a staged national model with a finer progression, so the level labels should be read as operationalized categories and not as the official classification of each entity.

External validity. The scope is limited to the Colombian public-sector X-Road context and to the entities registered in the central node; entities outside the integration process are not covered. The result is a time-bound snapshot as of the cut-off date, and the ecosystem evolves, so the diagnostic must be re-run to remain valid. No claim is made about other national interoperability architectures.

Conclusion validity. The study is descriptive; no experimental control group exists, so the relationship between the automated instrument and better decision-making (RQ4) is supported by the traceability of the results and by the elimination of manual valuation, not by a randomized trial. The automatic cross-check was computed over the seeded registry, whose size is smaller than the 127-entity population, so it should be read as a consistency check of the engine and not as a replication of the national diagnostic. Finally, no evaluation with real users of the interface was conducted in this study.

6 Security, Privacy, and Data Governance

The platform concentrates information about public institutions, their published services, their connection state, and the contact data of their technology officers. That combination is sensitive even though it is built from public sources: it enables systematic profiling of institutional weaknesses, and the CIO contact fields constitute personal data. The applicable regime is therefore the personal data protection law [1], together with the transparency and access to public information statute [2] that legitimizes the use of official public sources, and the regulatory framework that mandates the interoperability architecture [4]. The design principle adopted is *data minimization by construction*: the platform stores only the attributes required to compute the indicators, and every stored value is either institutional data or the professional contact data needed for fieldwork.

The controls implemented at the time of writing are the following. Access tokens are signed JWTs with the HS256 algorithm, issued with an access lifetime of 120 minutes and a refresh lifetime of 10 080 minutes; refresh tokens are typed and are rejected by the API if presented as access tokens. Passwords are never stored in clear text: they are hashed with bcrypt through Passlib, and the initial accounts are created by the API itself only when the user table is empty. Authorization is role-based, with an administrator role required for every write

operation on entities, services, and assessments, and an analyst role restricted to reading and analysis. Authentication is hardened against credential stuffing with a lockout rule: five consecutive failures suspend the account for fifteen minutes and return HTTP 423. Every relevant event—successful and failed logins, administrative actions, and access metadata with the client address—is written to an audit table in a non-blocking way, so that a failure while auditing can never break a user transaction. A global middleware enforces a per-address rate limit of 120 requests per minute and returns HTTP 429 beyond it, and a CORS allowlist restricts which origins may call the API. At the infrastructure level, the database is not published outside the container network, production connections require TLS, and configuration—including the signing secret—is injected through environment variables rather than committed to the repository. The interface handles the expiry of a token by attempting a refresh and, if that fails, by redirecting to authentication, which keeps the session model consistent across the application.

Two residual risks deserve explicit attention. First, the deployment uses free-tier managed services, so the platform inherits their availability profile and cannot offer the durability guarantees of an institutional hosting agreement; this is acceptable for a diagnostic instrument whose results are exported as artifacts, but not for a transactional citizen service. Second, the simulated connectors mean that no live data crosses the platform boundary today; the moment a real AND or MinTIC integration is enabled, an explicit data-processing instrument will be required before any transfer. Table 9 consolidates the requirements that must be closed before the instrument is used operationally.

The governance model that follows from these controls is simple: the platform stores evidence, not opinions. Each assessment records who produced it—the automatic engine or a named human assessor—when it was produced, which indicators contributed the points, and which recommendations were derived. Any third party with access to the deployed instrument and the exported reports can therefore recompute or contest a result, which is the property that distinguishes this diagnostic from a declarative compliance report.

7 Conclusions and Future Work

This article presented and evaluated a platform that automates the diagnosis of public-sector interoperability maturity in the Colombian X-Road ecosystem. Four conclusions follow from the design and the empirical results.

First, the automated instrument is technically feasible and produces a coherent diagnosis. The operationalization of the four MinTIC domains into explicit indicators, combined with a deterministic scoring rubric, yields a national maturity index of 2.3/5.0 at the *Basic* level, with the technical domain leading (3.1) and the semantic domain lagging (1.9). The cross-check performed by the automatic engine over the seeded registry reproduces the same ordering, which indicates that the result derives from the computation and not from a predefined narrative.

Table 9. Requirements pending before operational use of the platform.

Requirement	Rationale and required action
Replace simulated connectors with authenticated interfaces	Live data from the national systems requires a formal integration agreement and credential management; today the payloads are simulated.
Legal instrument for personal data processing	The CIO contact fields are personal data; a documented lawful basis, purpose limitation, retention policy, and registration of the database are required before processing at scale.
Managed secret management and rotation	The signing secret is injected through environment variables; it must be stored in a managed secret store and rotated periodically.
Alignment of token lifetimes with institutional policy	A 120-minute access token is convenient for analysis sessions but must be reviewed against the security policy of the adopting entity.
Encryption at rest and field-level protection	Institutional data currently relies on the managed database controls; field-level protection of contact data is recommended.
Independent penetration test	The hardening applied has not been validated by an external adversarial review.
Availability monitoring and alerting	The diagnostic should not depend on cold starts; uptime monitoring and a warm-up strategy are required for continuous use.
Longitudinal evidence retention	Assessments must be preserved under a formal retention and reproducibility policy to support multi-year studies.

Second, registration is not interoperability readiness. A third of the entities formally enrolled in the central node remain at the initial level, which means that an administrative catalog is a necessary but insufficient indicator of digital capability. Any public policy that measures progress by the number of registered members will overestimate the real state of the ecosystem.

Third, the binding constraint is semantic, not technological. The evidence shows that entities resolve connectivity before they resolve meaning: data dictionaries, metadata, schemas, and controlled vocabularies are the artifacts that are most frequently missing, and their absence forces bilateral mappings that do not scale. Consequently, investment in integration infrastructure without a parallel investment in semantic harmonization does not produce reusable interoperability.

Fourth, automated auditing reduces administrative burden while increasing traceability. Because every point is attributable to a named indicator and every assessment records its assessor and date, the instrument can be re-run periodically at marginal cost, which is precisely what a self-declared survey cannot do.

This supports a positive answer to RQ4, qualified by the limitations stated in Section 5.6.

Several lines of future work follow directly from the results. The most urgent one is to replace the simulated connectors with authenticated interfaces to the national systems, which requires an institutional agreement and will be the first step towards a production-grade diagnostic. In parallel, coverage should be extended to second- and third-level territorial entities, where the observed deficits are expected to be larger, and the semantic layer should be strengthened with concrete artifacts: published data dictionaries, reference schemas, and a conformance test suite that the platform can consume automatically.

Beyond the instrument itself, three research directions are proposed: (i) real-time availability monitoring for critical services, so that the diagnostic reflects operational continuity and not only declared availability; (ii) predictive analysis of citizen service demand based on data-consumption patterns, to anticipate the services that should be integrated first; and (iii) a longitudinal study of the evolution of the ecosystem, which the append-only assessment model already supports. Finally, a qualitative validation with the technology officers of the assessed entities—and a usability evaluation of the interface with real analysts—would close the remaining gap between an instrument that measures correctly and an instrument that is actually adopted by the institutions it is meant to serve. Comparative studies with other national X-Road implementations would additionally allow the Colombian results to be positioned against international references.

References

1. Congress of the Republic of Colombia: Law 1581 of 2012 — Provisions for the Protection of Personal Data. *Diario Oficial* 48.587 (2012)
2. Congress of the Republic of Colombia: Law 1712 of 2014 — Transparency and the Right of Access to National Public Information. *Diario Oficial* 49.084 (2014)
3. Congress of the Republic of Colombia: Law 1755 of 2015 — Regulation of the Fundamental Right of Petition. *Diario Oficial* 49.559 (2015)
4. President of the Republic of Colombia: Decree 1078 of 2015 — Single Regulatory Decree of the Information and Communications Technologies Sector (2015)
5. National Planning Department: CONPES 3975 — National Policy for Digital Transformation and Artificial Intelligence. DNP, Bogotá (2019)
6. Ministry of Information and Communications Technologies of Colombia: Interoperability Framework for Digital Government. MinTIC, Bogotá (2025)
7. Agencia Nacional Digital: Public Data and Documentation on Digital Interoperability. AND, Bogotá (2025)
8. Departamento Administrativo Nacional de Estadística: Directory of Public Entities of Colombia. DANE, Bogotá (2025)
9. Colombian Open Data Portal. <https://www.datos.gov.co> (2026), accessed September 2026
10. Nordic Institute for Interoperability Solutions: X-Road — Open-Source Data Exchange Layer for Interoperability. <https://x-road.global> (2026), accessed September 2026

11. European Commission: New European Interoperability Framework — Promoting Seamless Services and Data Flows for European Public Administrations. Publications Office of the European Union, Luxembourg (2017). <https://doi.org/10.2799/78681>
12. De Bruin, T., Freeze, R., Kulkarni, U., Rosemann, M.: Understanding the Main Phases of Developing a Maturity Assessment Model. In: Proceedings of the Australasian Conference on Information Systems (ACIS), pp. 8–19 (2005)
13. Becker, J., Knackstedt, R., Pöppelbuß, J.: Developing Maturity Models for IT Management. *Business & Information Systems Engineering* 1(3), 213–222 (2009). <https://doi.org/10.1007/s12599-009-0048-1>
14. Okan, A.A.: Exploring the Landscape of E-Government Maturity Models (2024)
15. Wahyuni, H.: Maturity of Data Interoperability Cross-Organizations in E-Government (2024)
16. Ramírez, S.: FastAPI — Modern, Fast Web Framework for Building APIs with Python. <https://fastapi.tiangolo.com> (2019), accessed September 2026
17. OpenAPI Initiative: OpenAPI Specification, version 3.1. <https://spec.openapis.org/oas/v3.1.0> (2021), accessed September 2026
18. PostgreSQL Global Development Group: PostgreSQL 15 Documentation. <https://www.postgresql.org/docs/15/> (2022), accessed September 2026
19. Meta Open Source: React — A JavaScript Library for Building User Interfaces. <https://react.dev> (2024), accessed September 2026
20. Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., et al.: Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research* 12, 2825–2830 (2011)
21. Docker Inc.: Docker Compose Documentation. <https://docs.docker.com/compose/> (2024), accessed September 2026